

Scholastic Aptitude Test, 2017 EC, entrance exam, booklet 395

Grade 12 · Aptitude · 2017 E.C.

Time allowed: 120 minutes

Questions: 55

Total marks: 55

Name _____ Grade _____ ID No. _____ Date _____

1. Gad is older than Dan and Dan is older than Betti. With Senni being older than Dan and younger than Gad, who is the youngest? (1)

A. Betti
B. Senni
C. Gad
D. Dan

2. Aki is Bali's sister. Yaya is Bali's mother. Lema is Yaya's father. Gette is Lema's mother. Then how is Aki related to Lema? (1)

A. daughter
B. granddaughter
C. grandson
D. grandmother

3. How many months have a twenty-eighth day? (1)

A. seven months
B. twelve months
C. one month
D. two months

4. *What is the matter with our reading is casualness, languor, preoccupation. We don't give the book a chance. We don't put ourselves at the disposal of the book. It is impossible to read properly without using all one's engine power. If we are not tired after reading, common sense is not in us. How one should grapple with a superior and not be out of breath? But even if we read with the whole force of our brain, and do nothing else, common sense is still not in us, while sublime conceit is. For we are assuming that, without further trouble, we can possess, co-ordinate, and assimilate all the ideas and sensations rapidly offered to us by a mind greater than our own. The assumption has only to be stated in order to appear in its monstrous absurdity. Hence, it follows that something remains to be done; This something is the act of reflection. Reading without subsequent reflection is ridiculous; it is equally a proof of folly and vanity. Source: Taken From: Warriner, J. E. et al. 1958. English Grammar & Composition. P420* (1)

What is described as an absurd assumption about reading according to the passage?

A. Reading does not require physical effort.
B. All books are equally valuable.
C. Ideas can be instantly absorbed and organized.
D. Superior minds are easy to understand.

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5. *What is the matter with our reading is casualness, languor, preoccupation. We don't give the book a chance. We don't put ourselves at the disposal of the book. It is impossible to read properly without using all one's engine power. If we are not tired after reading, common sense is not in us. How one should grapple with a superior and not be out of breath? But even if we read with the whole force of our brain, and do nothing else, common sense is still not in us, while sublime conceit is. For we are assuming that, without further trouble, we can possess, co-ordinate, and assimilate all the ideas and sensations rapidly offered to us by a mind greater than our own. The assumption has only to be stated in order to appear in its monstrous absurdity. Hence, it follows that something remains to be done; This something is the act of reflection. Reading without subsequent reflection is ridiculous; it is equally a proof of folly and vanity. Source: Taken From: Warriner, J. E. et al. 1958. English Grammar & Composition. P420* (1)

Which of the ideas is used as evidence for your answer to question number 7?

- A. Line 3-4 [If we are not ... is not in us.]
- B. Line 7-9 [For we ... greater than our own.]
- C. Line 4 [tired after reading... disposal of the book.]
- D. Line 1 [What is the ... preoccupation.]

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6. *What is the matter with our reading is casualness, languor, preoccupation. We don't give the book a chance. We don't put ourselves at the disposal of the book. It is impossible to read properly without using all one's engine power. If we are not tired after reading, common sense is not in us. How one should grapple with a superior and not be out of breath? But even if we read with the whole force of our brain, and do nothing else, common sense is still not in us, while sublime conceit is. For we are assuming that, without further trouble, we can possess, co-ordinate, and assimilate all the ideas and sensations rapidly offered to us by a mind greater than our own. The assumption has only to be stated in order to appear in its monstrous absurdity. Hence, it follows that something remains to be done; This something is the act of reflection. Reading without subsequent reflection is ridiculous; it is equally a proof of folly and vanity. Source: Taken From: Warriner, J. E. et al. 1958. English Grammar & Composition. P420* (1)

Why does the author emphasize the act of reflection?

- A. To complete the reading process.
- B. To simplify the ideas encountered.
- C. To improve intellectual superiority.
- D. To make reading more enjoyable.

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7. *What is the matter with our reading is casualness, languor, preoccupation. We don't give the book a chance. We don't put ourselves at the disposal of the book. It is impossible to read properly without using all one's engine power. If we are not tired after reading, common sense is not in us. How one should grapple with a superior and not be out of breath? But even if we read with the whole force of our brain, and do nothing else, common sense is still not in us, while sublime conceit is. For we are assuming that, without further trouble, we can possess, co-ordinate, and assimilate all the ideas and sensations rapidly offered to us by a mind greater than our own. The assumption has only to be stated in order to appear in its monstrous absurdity. Hence, it follows that something remains to be done; This something is the act of reflection. Reading without subsequent reflection is ridiculous; it is equally a proof of folly and vanity. Source: Taken From: Warriner, J. E. et al. 1958. English Grammar & Composition. P420* (1)

Which of the ideas is used as evidence for your answer to question number 9?

- A. Line 10 [The assumption... absurdity.]
- B. Line 12 [Reading without... ridiculous.]
- C. Line 4 [tired after reading... not in us.]
- D. Line 1 [What is the ... preoccupation.]

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- 8.** Choose the letter that contains the correct option. (1)
- A. If you'll not feel well, you'd better go to clinic.
 - B. If you're not feeling well, you'd better go to clinic.
 - C. If you'll not feel well, you better go to clinic.
 - D. If you're not feeling well, you would better go to clinic.
-
- 9.** Choose the letter that contains the correct option. (1)
- A. There school takes only twenty minutes on foot.
 - B. Theirs school takes only twenty minutes on foot.
 - C. Their school takes only twenty minutes on foot.
 - D. They're school takes only twenty minutes on foot.
-
- 10.** Choose the letter that contains the correct option. (1)
- A. I wanted to first take shower and then eat snack.
 - B. I wanted first take shower and then eat snack.
 - C. I wanted to first take shower and then ate snack.
 - D. I wanted to first take shower and then eating snack.
-
- 11.** Choose a word that does NOT fit to the group. (1)
- | | |
|-------------|--------------|
| A. Monogamy | B. Polygamy |
| C. Polyglot | D. Polyandry |
-
- 12.** Choose a word that does NOT fit to the group. (1)
- | | |
|---------------|---------------|
| A. Election | B. Ballot-box |
| C. Signal-box | D. Voting |
-
- 13.** Choose a word that does NOT fit to the group. (1)
- | | |
|--------------|-------------|
| A. Tiptoeing | B. Flying |
| C. Strolling | D. Striding |
-
- 14.** Choose a word that does NOT fit to the group. (1)
- | | |
|------------|---------|
| A. Yacht | B. Ship |
| C. Vehicle | D. Boat |
-
- 15.** AUTOMOBILE: GARAGE: : Hangar (1)
- | | |
|----------|-------------|
| A. Deck | B. Truck |
| C. Train | D. Airplane |
-
- 16.** KEY: LOCK:: Computer (1)
- | | |
|------------|-------------|
| A. Server | B. Screen |
| C. Monitor | D. Password |
-
- 17.** NEST: BIRD:: Lion (1)
- | | |
|-------------|---------|
| A. Backyard | B. Den |
| C. Hive | D. Cave |

-
- 18. PHYSICIAN: HOSPITAL:: School** (1)
- | | |
|------------------|------------|
| A. Administrator | B. Student |
| C. Supervisor | D. Teacher |
-
- 19. PHONEY** (1)
- | | |
|------------|--------------|
| A. Fake | B. Funny |
| C. Genuine | D. Authentic |
-
- 20. HUMOROUS** (1)
- | | |
|--------------|---------------|
| A. Satiating | B. Satisfying |
| C. Amusing | D. Doubting |
-
- 21. THEFT** (1)
- | | |
|-------------|--------------|
| A. Burglary | B. Hypocrisy |
| C. Gossip | D. Suspicion |
-
- 22. FONDLY** (1)
- | | |
|-----------------|--------------|
| A. Affectionate | B. Sociable |
| C. Hatred | D. Agreeable |
-
- 23. INVINCIBLE** (1)
- | | |
|--------------|----------------|
| A. Breakable | B. Debatable |
| C. Visible | D. Conquerable |
-
- 24. SUBORDINATE** (1)
- | | |
|-------------|-------------|
| A. Superior | B. Retainer |
| C. Soldier | D. Inferior |
-
- 25.** *If you want transition from being an employee to an employer, the first step is to serve your current employer as you would wish to be served. Most successful employers today have risen through the ranks, relying on their ability to manage and direct others effectively by suppressing the vices of employers. They started with no more opportunities than you have now. There are people in your community who could benefit from your skills and vice versa. For example, Mr. John Smith wants to sell his grocery store and start a movie theater, while someone else with a movie theater might want to trade for a grocery store. If you can connect these two individuals, you will serve both and earn a commission in the process. Source: Hill, N. (2003). The law of success: The 21st century edition (pp. 633-634)* (1)
- In Paragraph 2, the phrase these two individuals refers to
- | |
|---|
| A. John Smith & the person in movie industry. |
| B. the grocery store and the movie theatre. |
| C. the employer and the employee. |
| D. people and the community as a whole. |

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26. *If you want transition from being an employee to an employer, the first step is to serve your current employer as you would wish to be served. Most successful employers today have risen through the ranks, relying on their ability to manage and direct others effectively by suppressing the vices of employers. They started with no more opportunities than you have now. There are people in your community who could benefit from your skills and vice versa. For example, Mr. John Smith wants to sell his grocery store and start a movie theater, while someone else with a movie theater might want to trade for a grocery store. If you can connect these two individuals, you will serve both and earn a commission in the process. Source: Hill, N. (2003). The law of success: The 21st century edition (pp. 633-634)* (1)

What does the sentence most successful employers today have risen through the ranks, relying on their ability to manage and direct others effectively imply?

- A. Every successful employer is the result of his/her hard work and diligence.
- B. Having different ranks and rewards is the real manifestation of successful employers.
- C. All successful employers today have reached their level passing through lots of ups and downs.
- D. Most successful employers have proven their leadership ability through their journey.

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27. *If you want transition from being an employee to an employer, the first step is to serve your current employer as you would wish to be served. Most successful employers today have risen through the ranks, relying on their ability to manage and direct others effectively by suppressing the vices of employers. They started with no more opportunities than you have now. There are people in your community who could benefit from your skills and vice versa. For example, Mr. John Smith wants to sell his grocery store and start a movie theater, while someone else with a movie theater might want to trade for a grocery store. If you can connect these two individuals, you will serve both and earn a commission in the process. Source: Hill, N. (2003). The law of success: The 21st century edition (pp. 633-634)* (1)

The main intent of the writer in the text is to imply that

- A. the transition from employee to employer does not require much effort except having the interest.
- B. smooth transition from being an employer to employee is more of a matter of destiny than hard work.
- C. most successful employers are those who have achieved their dreams through trade-off.
- D. there are steps that one needs to undergo in the transition from employee to employer.

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28. *If you want transition from being an employee to an employer, the first step is to serve your current employer as you would wish to be served. Most successful employers today have risen through the ranks, relying on their ability to manage and direct others effectively by suppressing the vices of employers. They started with no more opportunities than you have now. There are people in your community who could benefit from your skills and vice versa. For example, Mr. John Smith wants to sell his grocery store and start a movie theater, while someone else with a movie theater might want to trade for a grocery store. If you can connect these two individuals, you will serve both and earn a commission in the process. Source: Hill, N. (2003). The law of success: The 21st century edition (pp. 633-634)* (1)

According to the passage, what does the word vices most likely denote?

- | | |
|---------------|--------------|
| A. deputies | B. strengths |
| C. weaknesses | D. ambitions |

-
29. The man was killed in a car accident. His body was dragged from the burning wreckage of his car. (1)

- | | |
|--------------|-----------|
| A. remains | B. fire |
| C. backseats | D. wheels |

-
30. Harvard is one of the most prestigious universities in the USA. (1)

- | | |
|---------------|--------------|
| A. privileged | B. admired |
| C. furnished | D. expensive |

-
31. The government has just repealed the law segregating public facilities so that there will be no longer discrimination. (1)
- A. approved
C. signed
- B. renewed
D. cancelled
-
32. Which one of the following statements is true? (1)
- A. Every integer number can be numerator of a rational number.
B. Every non-negative integer is a natural number.
C. The intersection of the set of integers and the set of irrational numbers is non-empty set.
D. The sum of two irrational numbers is irrational number.
-
33. If $f(x) = \sin \pi x + e^{x^2} + |x| + x^3$ and $g(x) = \cos \pi x + \ln(x + 1) + x$, then what is the value of $f(1) - 2g(0)$? (1)
- A. e
C. 0
- B. $\ln 2$
D. $e + 2$
-
34. A person makes a glass of juice and used half of the glass with avocado, one third of the remaining half with mangoes and the rest is filled by banana and papaya with the same proportion. Which one of the following is the ratio of bananas to avocados in the glass? (1)
- A. 1:6
C. 1:3
- B. 1:2
D. 2:3
-
35. Which one of the following is the largest number? (1)
- A. $\sqrt[6]{5}$
C. $\sqrt[12]{11}$
- B. $\sqrt[36]{2}$
D. $\sqrt[64]{3}$
-
36. Two pipes A and B are used to fill a water tank. Pipe A can fill the tank in 3 hours, and pipe B can fill the same tank in 6 hours. How long will it take in hours to fill the tank if both pipes are opened at the same time? (1)
- A. 5
C. 9
- B. 4
D. 2
-
37. A milk factory has only two pipe machines used to fill bottles for market. The old and new machines can fill 35 and 50 bottles per hour, respectively, and a total of 590 bottles are filled every day. One day the old machine was broken after working for 4 hours. How long will it take in hours for the new machine to fill the remaining bottles on that day? (1)
- A. 8
C. 13
- B. 5
D. 7
-
38. If a spherical soccer ball has a volume of $36\pi \text{ cm}^3$, then which one of the following is the surface area of the ball in square centimeter? (1)
- A. $\frac{16\pi}{3}$
C. 36π
- B. 12π
D. $\frac{32\pi}{3}$
-
39. A ball is shot into the air from the top of the building. The height in meter is given by $h(t) = kt - t^2 + 30$, where t is the time taken in second. What is the possible value of k if the maximum value of height from the ground is 34? (1)
- A. $\sqrt{2}$
C. 5
- B. 4
D. 2

40. A farmer has ten cows. Every day, he receives the same amount of milk in liters from each cow. One day, due to the food supply, he received half of the daily amount of milk from the four cows; he received three-fourth of the daily amount of milk from the three cows, and no milk from the other three cows. In total, he didn't receive ninety-two liters of milk on that day. How many liters of milk did he receive on that particular day? (1)
- A. 92
B. 16
C. 160
D. 68

41. For Questions 46 - 48, use the following information and answer each question based on the information therein. The following bar graph shows the data of a students' study pattern recorded for 32 days. (1)

How many days did the student study more than the average amount of the study?

[Figure: A photograph of a printed bar chart, slightly skewed, with faint show-through of text from the reverse of the page. Title above the plot: "Study Patterns of a Student". The vertical axis is labelled "Number of days" and is scaled 0 to 10 in steps of 1. The horizontal axis is labelled "Amount of study" and carries five categories, left to right: 2 hours, 4 hours, 6 hours, 8 hours, 10 hours. Five solid black bars, one per category, reach these heights: 2 hours = 10 days, 4 hours = 7 days, 6 hours = 7 days, 8 hours = 5 days, 10 hours = 3 days. The bar heights sum to the 32 days named in the instruction above the chart.]

- A. 5
B. 8
C. 17
D. 15

42. For Questions 46 - 48, use the following information and answer each question based on the information therein. The following bar graph shows the data of a students' study pattern recorded for 32 days. (1)

How many days did the student study less than 8 hours?

[Figure: A photograph of a printed bar chart, slightly skewed, with faint show-through of text from the reverse of the page. Title above the plot: "Study Patterns of a Student". The vertical axis is labelled "Number of days" and is scaled 0 to 10 in steps of 1. The horizontal axis is labelled "Amount of study" and carries five categories, left to right: 2 hours, 4 hours, 6 hours, 8 hours, 10 hours. Five solid black bars, one per category, reach these heights: 2 hours = 10 days, 4 hours = 7 days, 6 hours = 7 days, 8 hours = 5 days, 10 hours = 3 days. The bar heights sum to the 32 days named in the instruction above the chart.]

- A. 18
B. 6
C. 24
D. 12

43. For Questions 46 - 48, use the following information and answer each question based on the information therein. The following bar graph shows the data of a students' study pattern recorded for 32 days. (1)

Which time in hours is the median amount of the study?

[Figure: A photograph of a printed bar chart, slightly skewed, with faint show-through of text from the reverse of the page. Title above the plot: "Study Patterns of a Student". The vertical axis is labelled "Number of days" and is scaled 0 to 10 in steps of 1. The horizontal axis is labelled "Amount of study" and carries five categories, left to right: 2 hours, 4 hours, 6 hours, 8 hours, 10 hours. Five solid black bars, one per category, reach these heights: 2 hours = 10 days, 4 hours = 7 days, 6 hours = 7 days, 8 hours = 5 days, 10 hours = 3 days. The bar heights sum to the 32 days named in the instruction above the chart.]

- A. 6
B. 2
C. 8
D. 4

44. If $-x + 2y = 4$, then what is the value of $\frac{25^y}{5^x}$? (1)

- A. 125
B. 625
C. 5
D. 25

45. A woman burns 416 calories in an hour exercise at a fitness center. If she burns 10 calories per minute swimming in the water and 6 calories per minute pedaling on the stationary bike, then how many minutes of the hour does she spend exercising in the water? (1)
- A. 56
B. 14
C. 60
D. 46
-
46. Which one of the following is the domain of $g(x) = \ln(1 - x) + \ln x$? (1)
- A. $(0, \infty)$
B. $(\mathbb{R}) \setminus \{-1, 0\}$
C. $(0, 1)$
D. $(1, \infty)$
-
47. How many days are there from the first day of March to the last day of May? (1)
- A. 89
B. 92
C. 91
D. 90
-
48. A corporation pays ten employees a total of Birr 100,000 every month. Six out of ten employees receives less than Birr 10,000 per month. Quantity P: The average of the monthly payment of the corporation Quantity Q: The median of the monthly payment of the corporation (1)
- A. Quantity P is greater than Quantity Q.
B. Quantity P is less than Quantity Q.
C. The two quantities cannot be compared.
D. The two quantities are equal.
-
49. Consider the lengths of the sides of a triangle x, y, z , and the quantities given below. Quantity P: $3x + y - z$ Quantity Q: $4x$ (1)
- A. Quantity P is less than Quantity Q.
B. Quantity P is greater than Quantity Q.
C. The two quantities cannot be compared.
D. The two quantities are equal.
-
50. Which one of the following is true about the function $f(x) = x^{2/5}$? (1)
- A. The domain of f is $\{x \in \mathbb{R} \mid x \geq 0\}$
B. The range of f is $\{y \in \mathbb{R} \mid y \geq 0\}$
C. The range of f is $\{y \mid y \in \mathbb{R}\}$
D. The domain of f is $\{x \in \mathbb{R} \mid x \leq 0\}$
-
51. What is the value of z in the sequence 1, 2, 4, 8, 10, 20, 22, $z, \dots$? (1)
- A. 44
B. 24
C. 32
D. 40
-
52. An engineer is putting black and white colored ceramics on a square floor of area $36m^2$. Each ceramic has area $1m^2$. The engineer places the black ones along both diagonals and the rest is covered by white colored ceramics. In what ratio is black to white ceramic used? (1)
- A. 1:6
B. 1:2
C. 1:5
D. 4:5
-
53. A daily profit earned by a T-shirt factory is modeled by the formula $p(x) = -x^2 + 1,400x - 80,000$, where p is the profit and x is the price for each T-shirt sold. What range of prices will yield a profit of at least Birr 400,000? (1)
- A. $600 \leq x \leq 800$
B. $x \leq 800$
C. $x \leq 600$ or $x \geq 800$
D. $x \leq 600$

54. Points A, B and C are located on a circle with center O and radius 2 units long. If A, B and C are joined to form an equilateral triangle, then what is the area of triangle ABC in square units? (1)

A. 3

B. $\sqrt{3}$

C. 1

D. $3\sqrt{3}$

55. What is the possible value of the missing term x in the sequence -1, 0, 1, 2, -1, 8, 1, 26, -1, x, 1, 242, -1, 728, 1,...? (1)

A. 54

B. 36

C. 80

D. 34

— END OF EXAMINATION —

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